In [30]:
```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
```

In [27]:
```
import pandas as pd

fact = pd.read_csv("fact_lpg_shortage.csv")
dim_supplier = pd.read_csv("dim_supplier.csv")
dim_time = pd.read_csv("dim_time.csv")
dim_location = pd.read_csv("dim_location.csv")
dim_customer = pd.read_csv("dim_customer.csv")
```

In [28]:
```
df = fact.merge(dim_supplier, on="supplier_id") \
         .merge(dim_time, on="time_id") \
         .merge(dim_location, on="location_id") \
         .merge(dim_customer, on="customer_id")
```

In [29]:
```
df.info()
```

```
<class 'pandas.DataFrame'>
RangeIndex: 20000 entries, 0 to 19999
Data columns (total 34 columns):
 #   Column                 Non-Null Count  Dtype
---  ------                 --------------  -----
 0   transaction_id         20000 non-null  int64
 1   location_id            20000 non-null  int64
 2   supplier_id            20000 non-null  int64
 3   time_id                20000 non-null  int64
 4   customer_id            20000 non-null  int64
 5   demand_cylinders       20000 non-null  int64
 6   supply_cylinders       20000 non-null  int64
 7   shortage_cylinders     20000 non-null  int64
 8   price_per_cylinder     20000 non-null  float64
 9   subsidy_amount         20000 non-null  float64
 10  delivery_delay_days    20000 non-null  int64
 11  stock_available        20000 non-null  int64
 12  transport_cost         20000 non-null  float64
 13  weather_impact_score   20000 non-null  float64
 14  strike_impact_flag     20000 non-null  int64
 15  festival_demand_flag   20000 non-null  int64
 16  emergency_supply_flag  20000 non-null  int64
 17  warehouse_capacity     20000 non-null  int64
 18  fulfilled_orders       20000 non-null  int64
 19  unfulfilled_orders     20000 non-null  int64
 20  supplier_name          20000 non-null  str
 21  supplier_type          20000 non-null  str
 22  rating                 20000 non-null  float64
 23  date                   20000 non-null  str
 24  day                    20000 non-null  int64
 25  month                  20000 non-null  int64
 26  quarter                20000 non-null  int64
 27  year                   20000 non-null  int64
 28  city                   20000 non-null  str
 29  state                  20000 non-null  str
 30  region                 20000 non-null  str
 31  customer_type          20000 non-null  str
 32  income_group           20000 non-null  str
 33  connection_type        20000 non-null  str
dtypes: float64(5), int64(20), str(9)
memory usage: 5.2 MB
```

In [32]:
```
fact = fact.drop_duplicates()
```

In [33]:
```
fact[fact.duplicated()]
```

Out[33]:

transaction_id	location_id	supplier_id	time_id	customer_id	demand_cylinder

In [47]:
```
fact['subsidy_amount'] = fact['subsidy_amount'].fillna(0)
```

In [48]:
```
fact['shortage_cylinders'] = fact['shortage_cylinders'].astype(int)
fact['price_per_cylinder'] = fact['price_per_cylinder'].astype(float)
```

In [39]:
```
high_shortage = fact[fact['shortage_cylinders'] > 10]
```

In [40]:
```
fact['shortage_cylinders'] = pd.to_numeric(fact['shortage_cylinders'], errors=
```

In [41]:
```
total_shortage = fact['shortage_cylinders'].sum()
print(total_shortage)
```

```
190680
```

In [43]:
```
avg_delay = fact['delivery_delay_days'].mean()
print(avg_delay)
```

```
4.47975
```

In [44]:
```
df = fact.merge(dim_location, on="location_id")

state_shortage = df.groupby('state')['shortage_cylinders'].sum().reset_index()
```

In [ ]:
```
total_demand = df['demand_cylinders'].sum()
```

In [52]:
```
shortage_percent = (total_shortage / total_demand) * 100
print(shortage_percent)
```

```
38.07279915340535
```

In [53]:
```
state_shortage = df.groupby('state')['shortage_cylinders'].sum()
print(state_shortage)
```

```
state
State_0    18626
State_1    18153
State_2    19087
State_3    19471
State_4    19102
State_5    19079
State_6    19211
State_7    19129
State_8    19197
State_9    19625
Name: shortage_cylinders, dtype: int64
```

In [56]:
```
df = fact.merge(dim_supplier, on="supplier_id", how="left")
```

In [57]:
```
supplier_delay = df.groupby('supplier_name')['delivery_delay_days'].mean()
print(supplier_delay)
```

```
supplier_name
Supplier_1     4.602116
Supplier_10    4.351670
Supplier_11    4.337782
Supplier_12    4.500000
Supplier_13    4.556298
Supplier_14    4.458547
Supplier_15    4.524355
Supplier_16    4.511693
Supplier_17    4.551619
Supplier_18    4.413828
Supplier_19    4.505528
Supplier_2     4.457557
Supplier_20    4.444785
Supplier_3     4.445674
Supplier_4     4.300099
Supplier_5     4.539293
Supplier_6     4.435045
Supplier_7     4.440000
Supplier_8     4.614754
Supplier_9     4.602956
Name: delivery_delay_days, dtype: float64
```

In [60]:
```
df = fact.merge(dim_time, on="time_id", how="left")
```

In [64]:
```
df['date'] = pd.to_datetime(df['date'])

monthly_shortage = df.groupby(df['date'].dt.month)['shortage_cylinders'].sum()
```

In [65]:
```
df.head()
```

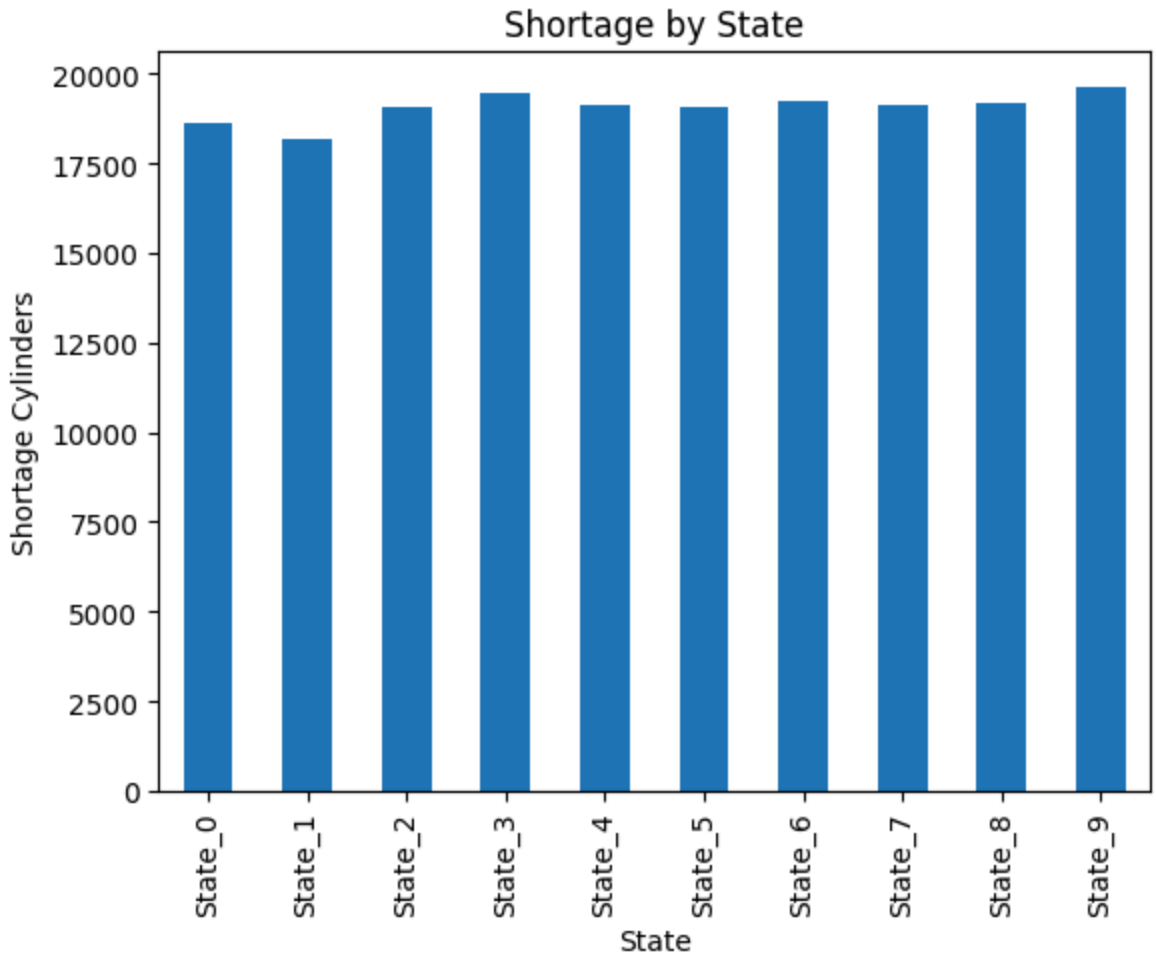
Out[65]:

	transaction_id	location_id	supplier_id	time_id	customer_id	demand_cylinde
0	1	48	2	121	34	
1	2	16	1	184	58	
2	3	3	10	48	82	
3	4	20	1	147	80	
4	5	24	6	282	61	

5 rows × 25 columns

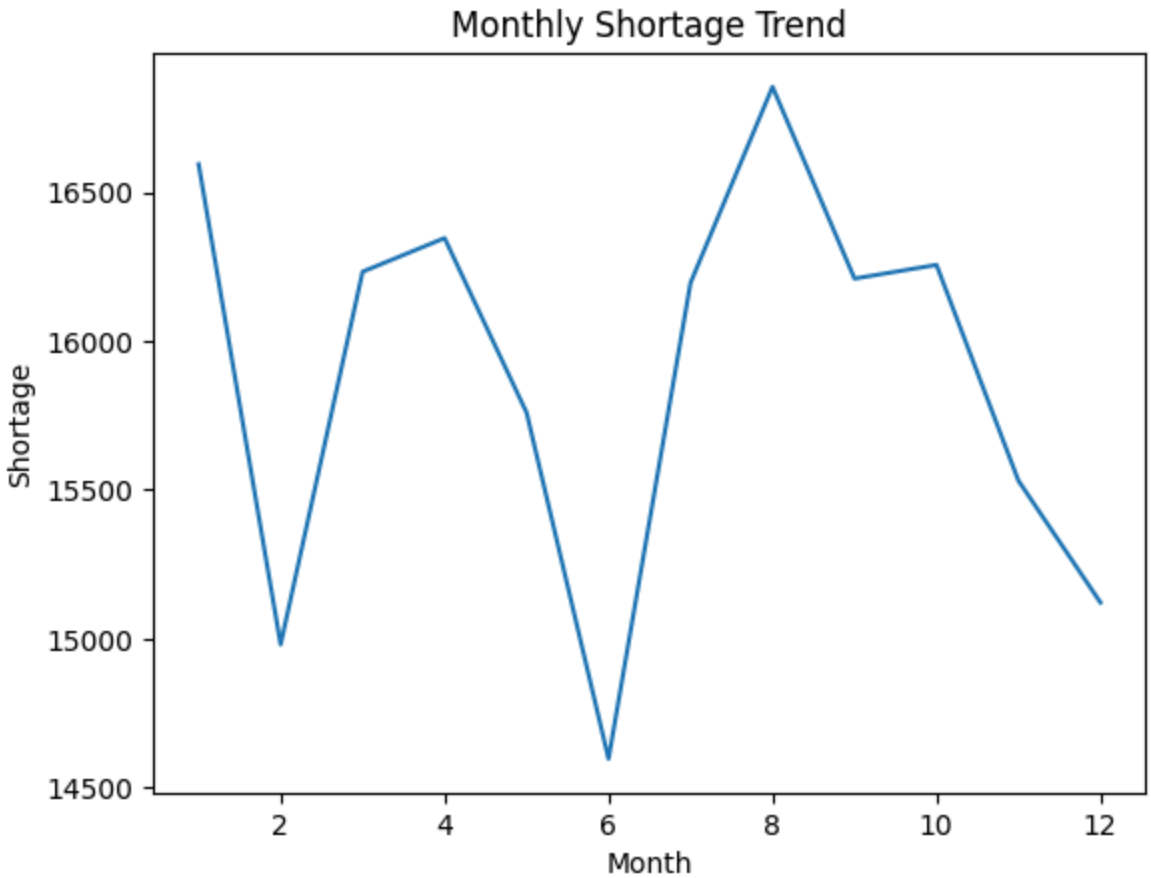
In [ ]:
```
##  1. Bar Chart – Shortage by State
```

In [66]:
```
state_shortage.plot(kind='bar')
plt.title("Shortage by State")
plt.xlabel("State")
plt.ylabel("Shortage Cylinders")
plt.show()
```



In [ ]:
```
##   2. Line Chart – Monthly Trend
```

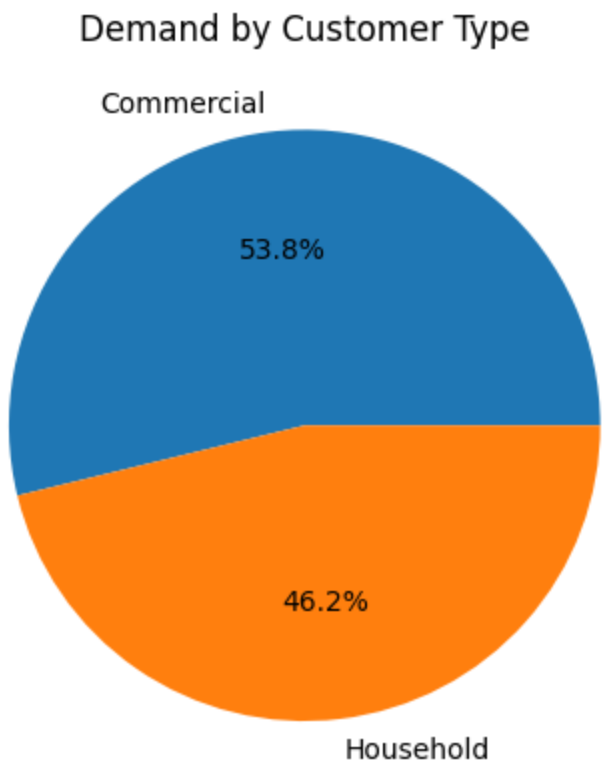
In [67]:
```
monthly_shortage.plot(kind='line')
plt.title("Monthly Shortage Trend")
plt.xlabel("Month")
plt.ylabel("Shortage")
plt.show()
```



In [69]:
```
df = fact.merge(dim_customer, on="customer_id", how="left")
```

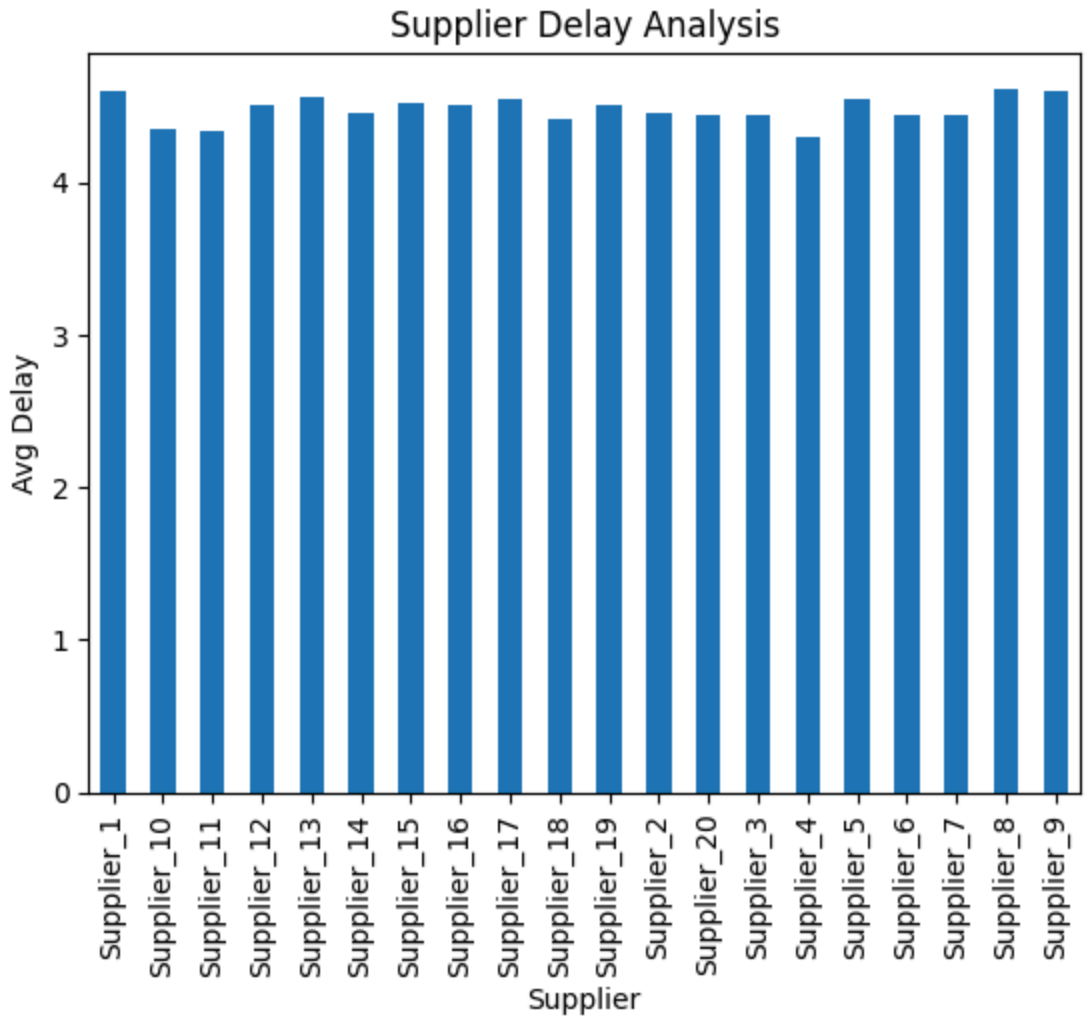
In [ ]:
```
## 3. Pie Chart – Customer Type
```

In [70]:
```
customer_demand = df.groupby('customer_type')['demand_cylinders'].sum()

customer_demand.plot(kind='pie', autopct='%1.1f%%')
plt.title("Demand by Customer Type")
plt.ylabel("")
plt.show()
```



In [ ]:
```
## 4. Supplier Performance
```

In [71]:
```
supplier_delay.plot(kind='bar')
plt.title("Supplier Delay Analysis")
plt.xlabel("Supplier")
plt.ylabel("Avg Delay")
plt.show()
```

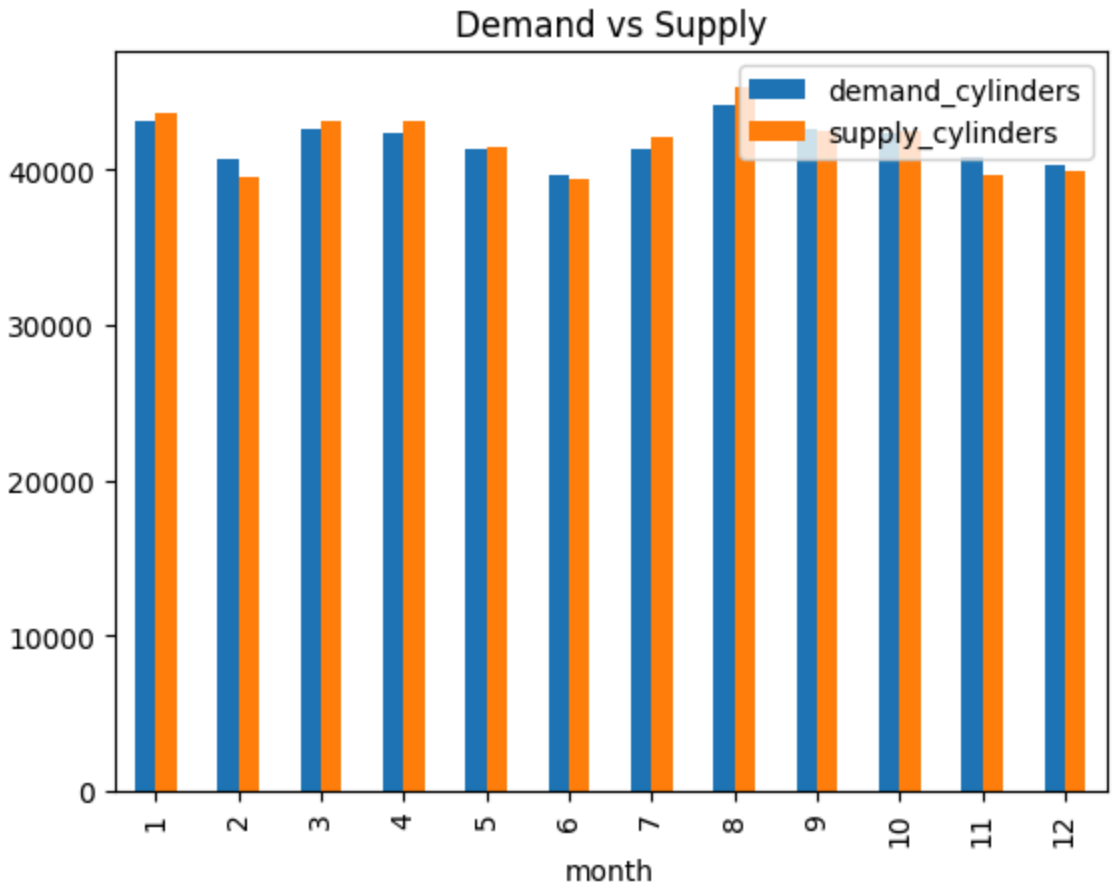


In [ ]:
```
##  5. Demand vs Supply
```

In [74]:
```
df = fact.merge(dim_time, on="time_id", how="left")
```

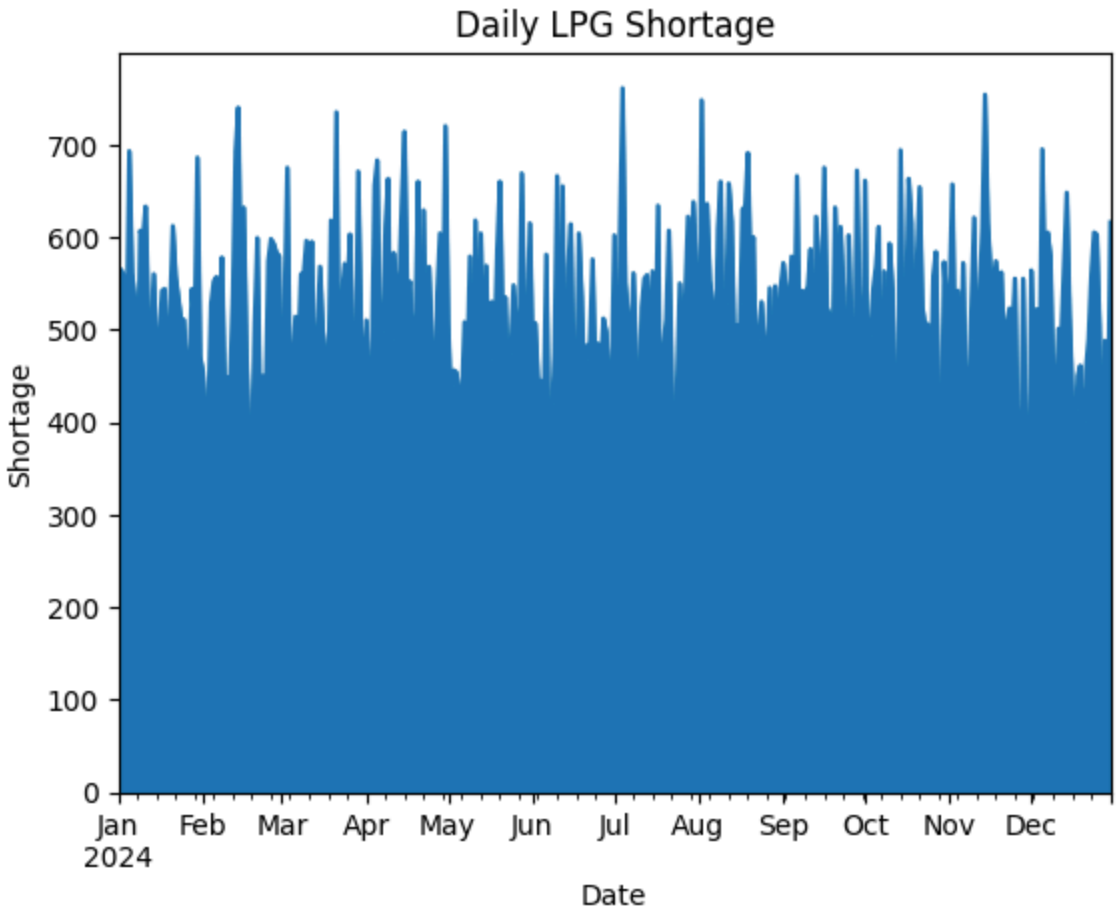
In [75]:
```
df['date'] = pd.to_datetime(df['date'])

df['month'] = df['date'].dt.month

ds = df.groupby('month')[['demand_cylinders','supply_cylinders']].sum()
```

In [76]:
```
ds = df.groupby('month')[['demand_cylinders','supply_cylinders']].sum()
ds.plot(kind='bar')
plt.title("Demand vs Supply")
plt.show()
```



In [ ]:
```
##  6. Area Chart – Daily Shortage
```

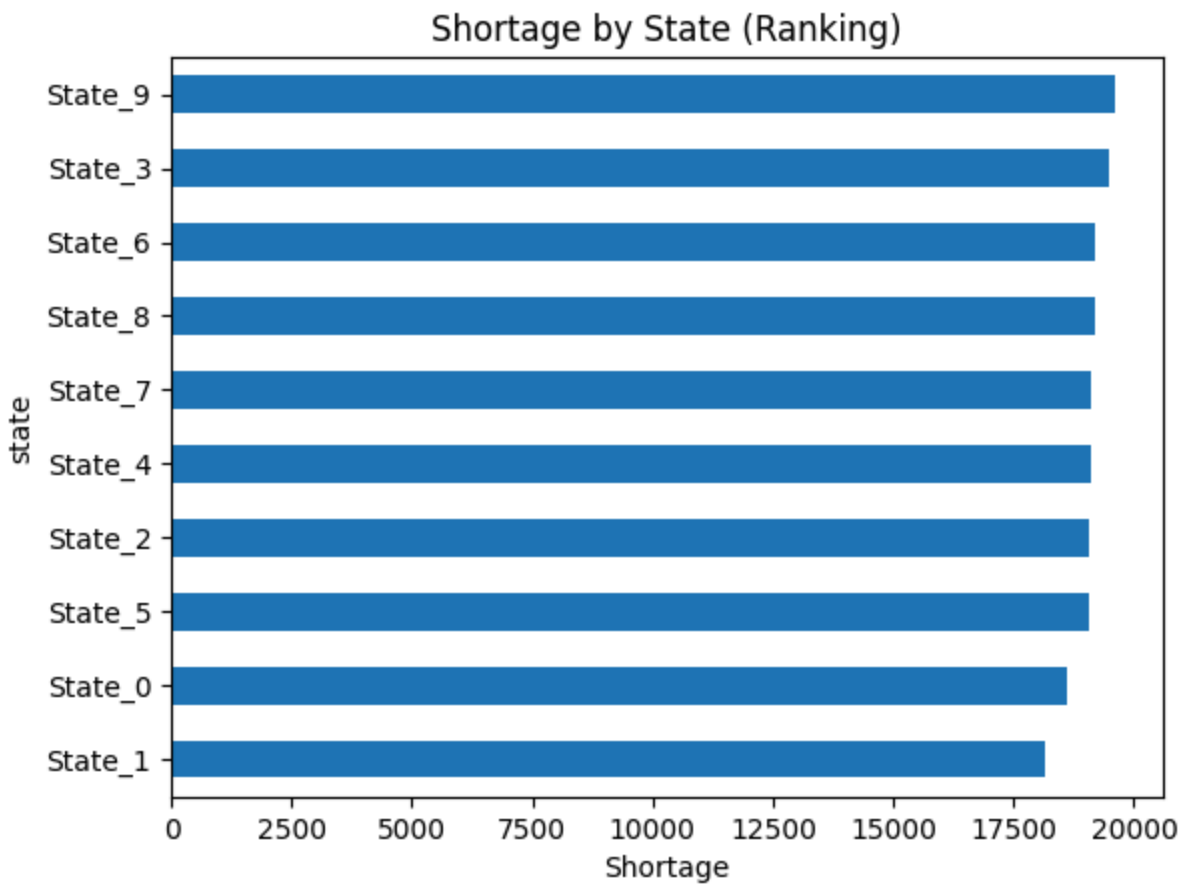
In [77]:
```
df['date'] = pd.to_datetime(df['date'])

daily_shortage = df.groupby('date')['shortage_cylinders'].sum()

daily_shortage.plot(kind='area')
plt.title("Daily LPG Shortage")
plt.xlabel("Date")
plt.ylabel("Shortage")
plt.show()
```



In [79]:
```
df = fact.merge(dim_supplier, on="supplier_id", how="left") \
         .merge(dim_time, on="time_id", how="left") \
         .merge(dim_location, on="location_id", how="left") \
         .merge(dim_customer, on="customer_id", how="left")
```

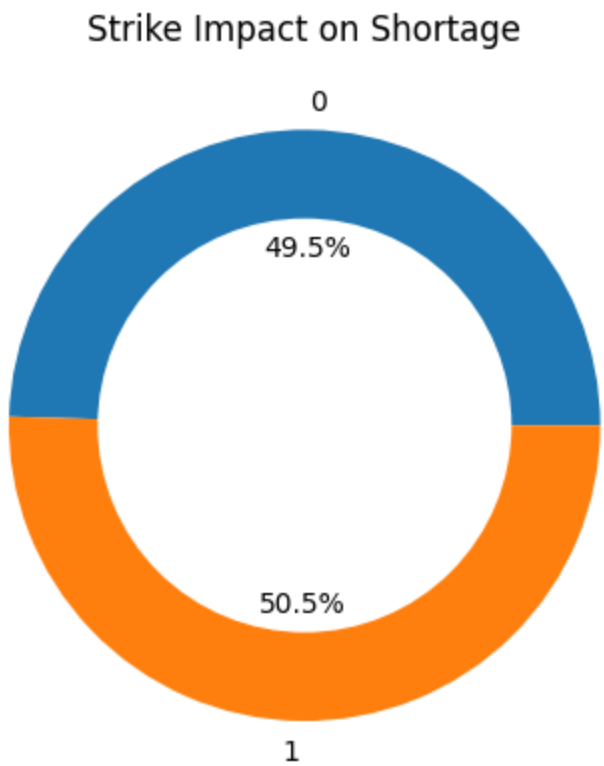
In [ ]:
```
##  7. Horizontal Bar – State Ranking
```

In [80]:
```
state_shortage = df.groupby('state')['shortage_cylinders'].sum().sort_values()

state_shortage.plot(kind='barh')
plt.title("Shortage by State (Ranking)")
plt.xlabel("Shortage")
plt.show()
```



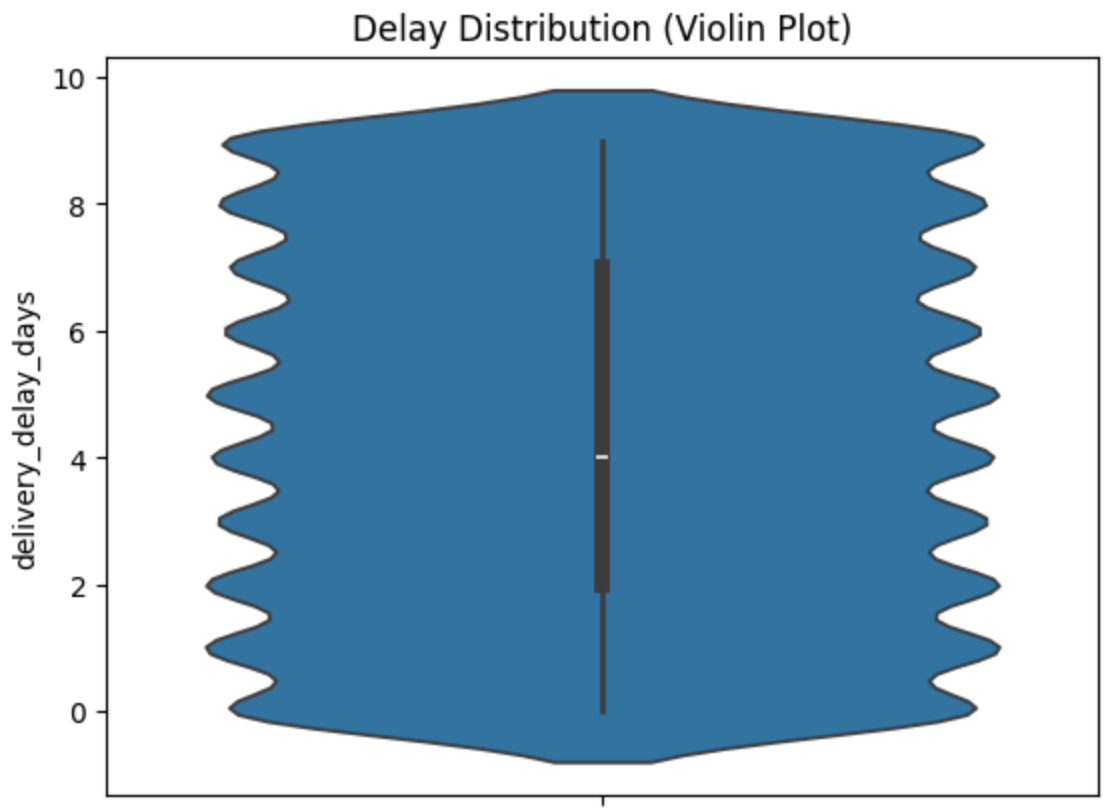
In [ ]:
```
##  8. Donut Chart – Strike Impact
```

In [81]:
```
strike = df.groupby('strike_impact_flag')['shortage_cylinders'].sum()

plt.pie(strike, labels=strike.index, autopct='%1.1f%%')
plt.title("Strike Impact on Shortage")

# donut effect
centre_circle = plt.Circle((0,0),0.70,fc='white')
fig = plt.gcf()
fig.gca().add_artist(centre_circle)

plt.show()
```



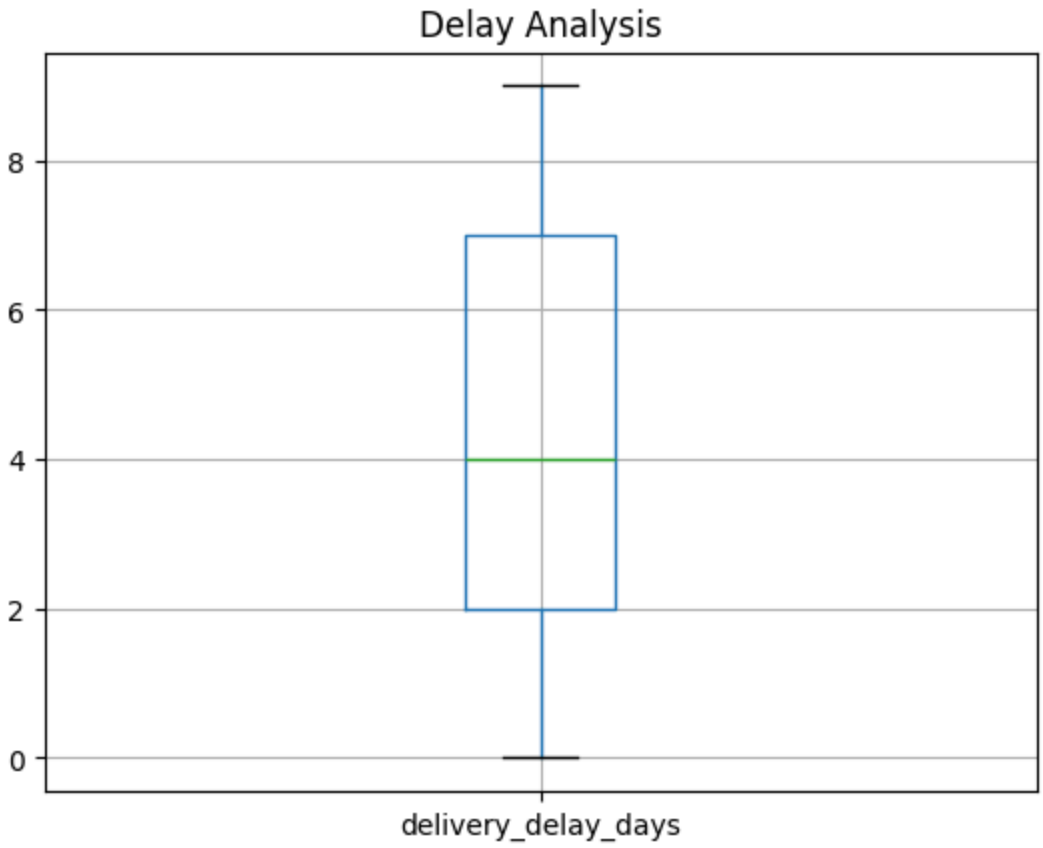
In [ ]:
```
## 9. VIOLIN PLOT Delay Distribution
```

In [84]:
```
import seaborn as sns

sns.violinplot(y=df['delivery_delay_days'])
plt.title("Delay Distribution (Violin Plot)")
plt.show()
```



In [ ]:
```
##  10. Box Plot – Delay Analysis
```

In [83]:
```
df.boxplot(column='delivery_delay_days')
plt.title("Delay Analysis")
plt.show()
```



In [ ]:

In [ ]:

In [ ]:

In [ ]: